

## Graphs of functions - Set 2

TMUA practice questions by topic. Specification MM8.1, MM8.5-MM8.7, M4.15. 6 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 The curves  $y = 2^x$  and  $y = x^2$  intersect at how many points?
- A 3
  - B 0
  - C 2
  - D 4
  - E 1
- 2 For how many values of  $k$  does the equation  $|x^2 - 4| = k$  have exactly three real solutions?
- A 1
  - B 0
  - C 2
  - D 3
  - E infinitely many
- 3 The graph of  $y = \sqrt{x}$  is transformed to  $y = \sqrt{4 - x}$ . Which sequence of transformations achieves this?
- A translate 4 units in the positive  $x$ -direction, then reflect in the  $y$ -axis
  - B translate 4 units in the negative  $x$ -direction, then reflect in the  $y$ -axis
  - C reflect in the  $x$ -axis, then translate 4 units in the positive  $y$ -direction
  - D stretch parallel to the  $x$ -axis with factor 4
  - E reflect in the  $y$ -axis, then translate 4 units in the positive  $x$ -direction

- 4 The cubic  $y = x^3 + ax + b$  has three distinct real roots. Which of the following statements must be true?

- I  $a < 0$
- II  $b \neq 0$
- III The curve has two turning points.

- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III

- 5 The graphs of  $y = f(x)$  and  $y = g(x)$  meet at exactly three points. Which of the following statements must be true?

- I The equation  $f(x) = g(x)$  has exactly three real solutions.
- II The graph of  $y = f(x) - g(x)$  crosses the  $x$ -axis exactly three times.
- III  $f(x) - g(x)$  is a polynomial of degree at least 3.

- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III

- 6**  $p(x)$  is a polynomial of degree 4 with real coefficients. Which of the following statements must be true?
- I**  $p(x) = 0$  has at least one real root.
  - II**  $p(x) = 0$  has at most four real roots.
  - III** The graph of  $y = p(x)$  has at least one turning point.
- A** none of them
  - B** I only
  - C** II only
  - D** III only
  - E** I and II only
  - F** I and III only
  - G** II and III only
  - H** I, II and III